SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA1402

COURSE NAME: Compiler Design for Higher

Level Processing

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error

recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS:4 Total Marks:50

Annexure A

DESIGN TASK

Code of Conduct:

I **_GAUTHAM B (192421318)_____** (Name / Reg No) certify that this submission is my original work and that I have

adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Course Faculty

GAUTHAM B

(192421318)

CSA1402 COMPILER DESIGN

PROGRAM 1:

```
#include <stdio.h>
#include <string.h>

char input[50];
char stack[50];
int top = -1;
int ip = 0;

void push(char c) {
    stack[++top] = c;
}

char pop() {
    return stack[top--];
}

char peek() {
    return stack[top];
}

void printStack() {
    for (int i = 0; i <= top; i++)
        printf("%c", stack[i]);
```

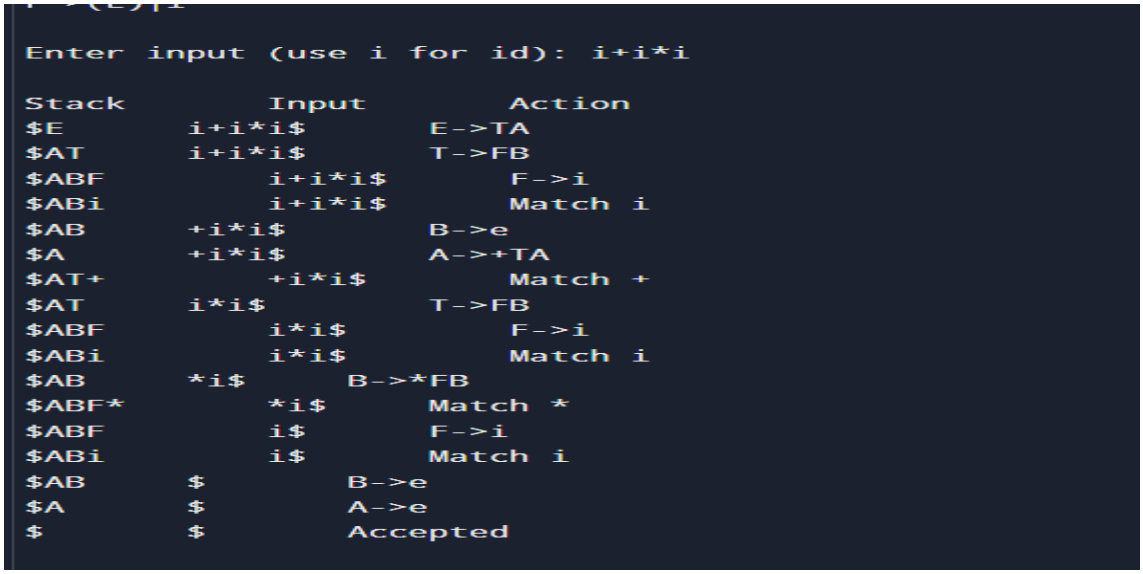
```
}

int main() {

    printf("Grammar:\n");
    printf("E->TA\n");
    printf("A->+TA|e\n");
    printf("T->FB\n");
    printf("B->*FB|e\n");
    printf("F->(E)|i\n\n");

    printf("Enter input (use i for id): ");
    scanf("%s", input);

    strcat(input, "$");

    push('$');
    push('E');

    printf("\nStack\t\tInput\t\tAction\n");

    while (top != -1) {

        printStack();
        printf("\t\t%s\t\t", input + ip);

        char x = peek();
        char a = input[ip];

        if (x == a && x == '$') {
            printf("Accepted\n");
            break;
        }
```

```
else if (x == 'i' || x == '+' || x == '*' || x == '(' || x == ')' || x == '$') {

    if (x == a) {
        pop();
        ip++;
        printf("Match %c\n", a);
    } else {
        printf("Error\n");
        break;
    }
}

else if (x == 'E') {
    if (a == 'i' || a == '(') {
        pop();
        push('A');
        push('T');
        printf("E->TA\n");
    } else {
        printf("Error\n");
        break;
    }
}

else if (x == 'A') {
    if (a == '+') {
        pop();
        push('A');
        push('T');
        push('+');
        printf("A->+TA\n");
    } else if (a == ')' || a == '$') {
```

```
        pop();
        printf("A->e\n");
    } else {
        printf("Error\n");
        break;
    }
}

else if (x == 'T') {
    if (a == 'i' || a == '(') {
        pop();
        push('B');
        push('F');
        printf("T->FB\n");
    } else {
        printf("Error\n");
        break;
    }
}

else if (x == 'B') {
    if (a == '*') {
        pop();
        push('B');
        push('F');
        push('*');
        printf("B->*FB\n");
    } else if (a == '+' || a == ')' || a == '$') {
        pop();
        printf("B->e\n");
    } else {
        printf("Error\n");
        break;
```

```
            }
        }

        else if (x == 'F') {
            if (a == 'i') {
                pop();
                push('i');
                printf("F->i\n");
            } else if (a == '(') {
                pop();
                push(')');
                push('E');
                push('(');
                printf("F->(E)\n");
            } else {
                printf("Error\n");
                break;
            }
        }
    }

    return 0;
}
```

Output:



PROGRAM 2:

```
#include <stdio.h>
#include <string.h>

char input[50], stack[50];
int top = -1, i = 0;

void printState(char action[]) {
    printf("%-15s %-15s %s\n", stack, input + i, action);
}

int main() {
    printf("Grammar:\n");
    printf("E -> E+E\n");
    printf("E -> E*E\n");
    printf("E -> i\n\n");

    printf("Enter input (use i for id): ");
    scanf("%s", input);

    strcat(input, "$");
```

```
printf("\n%-15s %-15s %s\n", "Stack", "Input", "Action");

while (1) {

    // Shift
    stack[++top] = input[i++];
    stack[top + 1] = '\0';
    printState("Shift");

    // Reduce i -> E
    if (top >= 0 && stack[top] == 'i') {
        stack[top] = 'E';
        printState("Reduce E->i");
    }

    // Reduce E*E -> E (Higher precedence)
    while (top >= 2 &&
           stack[top] == 'E' &&
           stack[top - 1] == '*' &&
           stack[top - 2] == 'E') {

        top -= 2;
        stack[top] = 'E';
        stack[top + 1] = '\0';
        printState("Reduce E->E*E");
    }

    // Reduce E+E -> E
    while (top >= 2 &&
           stack[top] == 'E' &&
           stack[top - 1] == '+' &&
           stack[top - 2] == 'E') {
```

```
    // Delay reduction if next input is '*'
    if (input[i] == '*')
        break;

    top -= 2;
    stack[top] = 'E';
    stack[top + 1] = '\0';
    printState("Reduce E->E+E");
}

// Accept
if (strcmp(stack, "E") == 0 && input[i] == '$') {
    printState("Accept");
    printf("\nString Accepted\n");
    break;
}

if (input[i] == '$' && strcmp(stack, "E") != 0) {
    while (top >= 2 &&
           stack[top] == 'E' &&
           stack[top - 1] == '+' &&
           stack[top - 2] == 'E') {

        top -= 2;
        stack[top] = 'E';
        stack[top + 1] = '\0';
        printState("Reduce E->E+E");
    }

    if (strcmp(stack, "E") == 0) {
        printState("Accept");
        printf("\nString Accepted\n");
```

```
        } else {
            printf("\nString Rejected\n");
        }
        break;
    }
  }

  return 0;
}
```

OUTPUT:

```
Enter input (use i for id): i+i*i

Stack           Input           Action
i               +i*i$           Shift
E               +i*i$           Reduce E->i
E+              i*i$            Shift
E+i             *i$             Shift
E+E             *i$             Reduce E->i
E+E*            i$              Shift
E+E*i           $               Shift
E+E*E           $               Reduce E->i
E+E             $               Reduce E->E*E
E               $               Reduce E->E+E
E               $               Accept

String Accepted
```

PROGRAM 3:

```
#include <stdio.h>
#include <string.h>
char input[100];
char stack[100];
int top = -1, ip = 0;
void push(char x)
{
  stack[++top] = x;
```

```
}
void pop()
{
    top--;
}
void displayStack()
{
    for (int i = 0; i <= top; i++)
        printf("%c", stack[i]);
}

int main()
{
    printf("Grammar after removing Left Recursion:\n");
    printf("S -> TA\n");
    printf("A -> +TA | e\n");
    printf("T -> FB\n");
    printf("B -> *FB | e\n");
    printf("F -> (S) | i\n\n");

    printf("Enter input (use i for id): ");
    scanf("%s", input);

    strcat(input, "$");

    push('$');
    push('S');

    printf("\n%-15s %-15s %s\n", "Stack", "Input", "Action");

    while (top >= 0)
    {
        displayStack();
```

```
printf("\t\t%s\t\t", input + ip);

char X = stack[top];
char a = input[ip];

// Accept
if (X == '$' && a == '$')
{
   printf("Accepted\n");
   printf("\nString Accepted\n");
   break;
}

// Match terminals
if (X == 'i' || X == '+' || X == '*' || X == '(' || X == ')' || X == '$')
{
   if (X == a)
   {
      pop();
      ip++;
      printf("Match %c\n", a);
   }
   else
   {
      printf("Error\n");
      break;
   }
}

// S -> TA
else if (X == 'S')
{
   if (a == 'i' || a == '(')
```

```
    {
        pop();
        push('A');
        push('T');
        printf("S->TA\n");
    }
    else
    {
        printf("Error\n");
        break;
    }
}

// A -> +TA | e
else if (X == 'A')
{
    if (a == '+')
    {
        pop();
        push('A');
        push('T');
        push('+');
        printf("A->+TA\n");
    }
    else if (a == ')' || a == '$')
    {
        pop();
        printf("A->e\n");
    }
    else
    {
        printf("Error\n");
        break;
```

```
    }
}

// T -> FB
else if (X == 'T')
{
    if (a == 'i' || a == '(')
    {
        pop();
        push('B');
        push('F');
        printf("T->FB\n");
    }
    else
    {
        printf("Error\n");
        break;
    }
}

// B -> *FB | e
else if (X == 'B')
{
    if (a == '*')
    {
        pop();
        push('B');
        push('F');
        push('*');
        printf("B->*FB\n");
    }
    else if (a == '+' || a == ')' || a == '$')
    {
```

```
        pop();
        printf("B->e\n");
    }
    else
    {
        printf("Error\n");
        break;
    }
}

// F -> i | (S)
else if (X == 'F')
{
    if (a == 'i')
    {
        pop();
        push('i');
        printf("F->i\n");
    }
    else if (a == '(')
    {
        pop();
        push(')');
        push('S');
        push('(');
        printf("F->(S)\n");
    }
    else
    {
        printf("Error\n");
        break;
    }
}
```

```
    }

    return 0;
}
```

OUTPUT:

```
Enter input (use i for id): i+i*i

Stack            Input            Action
$S       i+i*i$       S->TA
$AT      i+i*i$       T->FB
$ABF         i+i*i$       F->i
$ABi         i+i*i$       Match i
$AB      +i*i$        B->e
$A       +i*i$        A->+TA
$AT+         +i*i$        Match +
$AT      i*i$         T->FB
$ABF         i*i$         F->i
$ABi         i*i$         Match i
$AB      *i$      B->*FB
$ABF*        *i$      Match *
$ABF         i$       F->i
$ABi         i$       Match i
$AB      $        B->e
$A       $        A->e
$        $        Accepted

String Accepted
```

PROGRAM 4: